14	The piston is pushed very slowly into a metal cylinder containing an ideal gas. If the temperature of the gas does not change in the process. Which of the following statements is ***incorrect***?	
	A	The work done on the gas is equal to the heat lost by the gas.
	B	The pressure of the gas increases.
	C	The average kinetic energy of the gas molecules increases due to the work done on the gas.
	D	The internal energy of the gas remains constant.
	Answer: C **Metal → good conductor of heat. There can be heat transfer in/out of the cylinder.** **A: $\Delta U = \frac{3}{2}nR\Delta T = 0$. Hence by First Law of thermodynamics, W = -Q. Correct.**	

	B: Ideal gas equation, pV = nRT = constant since T & n are both constants. So since V decreases, p increases. Correct C: $\langle E_K \rangle = \frac{3}{2}kT$. Same temperature means that the mean translational KE is the same. Incorrect D: internal energy of an ideal gas = 3/2 nRT. For a fixed n, internal energy remains constant at constant T.